

# TokTickIT

Requester-facing ticket workflow

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| Student             | GUNTEE DOUNGMANEE                                               |
| Repository          | <a href="#">ovenmakemeheat/tokticket</a>                        |
| Report Issue        | <a href="#">Issue #65 - Lab 2 final submission report</a>       |
| Integrated baseline | lab2-staging at 83da75e                                         |
| Stable branch       | main at b9903e8 (Lab 1 release)                                 |
| Stack               | React, TypeScript, Vite, Bootstrap, Express, Prisma, PostgreSQL |

**DRAFT - 2026-09-04.** Lab 2 implementation and recorded evidence are integrated on **lab2-staging**. The release PR to final **main**, final-main test output, human visual sign-off, and several submission captures are still pending.

## Contents

|          |                                                                         |          |
|----------|-------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Answer Part 1: Git Use with Engineering Workflow</b>                 | <b>2</b> |
| <b>2</b> | <b>Answer Part 2: Spec DD</b>                                           | <b>3</b> |
| <b>3</b> | <b>Answer Part 3: Test DD and Traceability</b>                          | <b>3</b> |
| <b>4</b> | <b>Answer Part 4: AI Use with Reflection</b>                            | <b>4</b> |
| <b>5</b> | <b>Answer Part 5: Development Requester Select Screen</b>               | <b>4</b> |
| <b>6</b> | <b>Answer Part 6: Working Ticket Screen - Create Mode</b>               | <b>5</b> |
| <b>7</b> | <b>Answer Part 7: Working My Tickets Screen</b>                         | <b>7</b> |
| <b>8</b> | <b>Answer Part 8: Working Ticket Screen - View Mode and Attachments</b> | <b>7</b> |
| <b>9</b> | <b>Answer Part 9: Zen Green UI and Responsive Evidence</b>              | <b>8</b> |

# 1 Answer Part 1: Git Use with Engineering Workflow

Lab 2 uses one feature branch per Issue and integrates each reviewed Pull Request into `lab2-staging`. The human reviewer merges the feature PR; the PR author does not merge it. The current repository state is:

```
main (b9903e8, Lab 1 release)
|-- lab2-staging (83da75e)
|   |-- feature/5-sprint-specification    (#51 -> PR #58)
|   |-- feature/6-requester-context      (#52 -> PR #59)
|   |-- feature/7-create-ticket          (#53 -> PR #60)
|   |-- feature/8-my-tickets              (#54 -> PR #61)
|   |-- feature/9-ticket-detail           (#55 -> PR #62)
|   |-- feature/10-zen-green              (#56 -> PR #63)
|   |-- feature/11-e2e-evidence           (#57 -> PR #64)
|   |-- feature/12-report-compliance      (#65, active report branch)
```

| Issue | Feature branch                     | PR  | Current state                          |
|-------|------------------------------------|-----|----------------------------------------|
| #51   | feature/<br>5-sprint-specification | #58 | Merged into lab2-staging by MadMax168. |
| #52   | feature/6-requester-context        | #59 | Merged into lab2-staging by MadMax168. |
| #53   | feature/7-create-ticket            | #60 | Merged into lab2-staging by MadMax168. |
| #54   | feature/8-my-tickets               | #61 | Merged into lab2-staging by MadMax168. |
| #55   | feature/9-ticket-detail            | #62 | Merged into lab2-staging by MadMax168. |
| #56   | feature/10-zen-green               | #63 | Merged into lab2-staging by MadMax168. |
| #57   | feature/11-e2e-evidence            | #64 | Merged into lab2-staging by MadMax168. |

The complete review index and response summaries are in [docs/lab-02/reviewer.md](#). The record identifies MadMax168 as the human reviewer and records the requested changes and author responses for PRs #59, #61, #62, and #64. GitHub currently exposes human ready-to-merge or approve comments, but its formal `reviewDecision` is empty for PRs #58-#64; the report must not label those comments as formal **APPROVED** reviews. The project-board state also needs a human capture; the current CLI token cannot read the board without the `read:project` scope.

## Required Part 1 captures

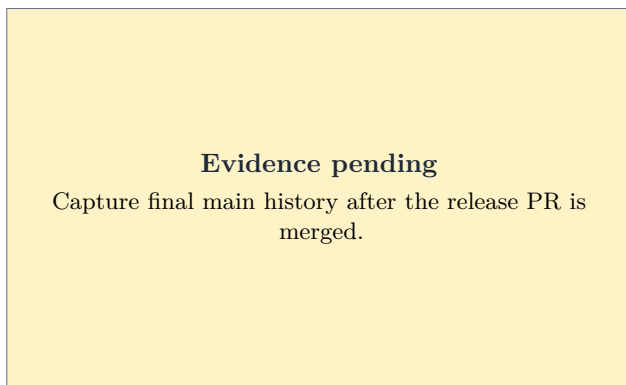


Figure 1: Final `main` history showing feature branches, staging merges, and the release merge.

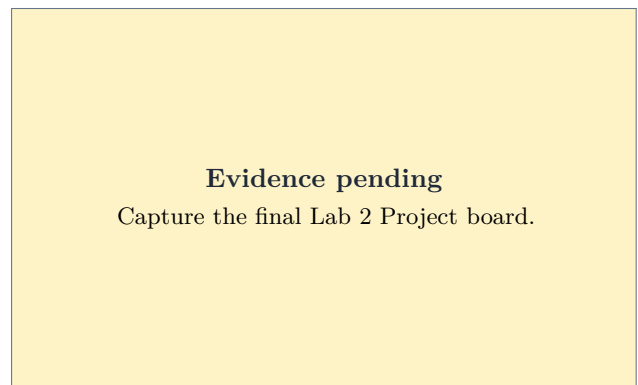


Figure 2: Final Kanban with the Lab 2 Issues in Done.

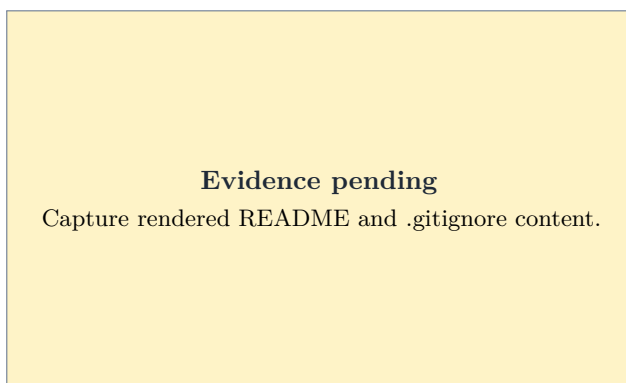


Figure 3: README and `.gitignore` evidence.

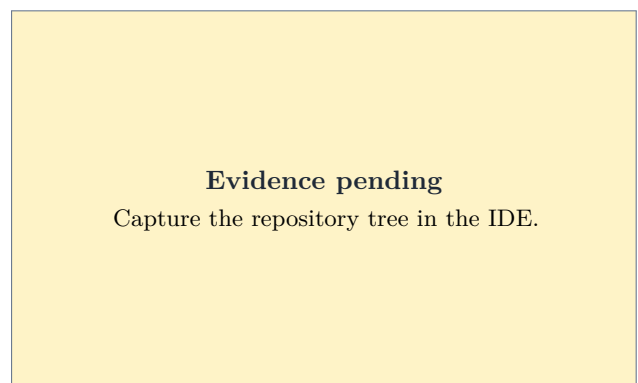


Figure 4: Repository directory structure.

## 2 Answer Part 2: Spec DD

The complete contract is linked at [docs/lab-02/specification.md](#) and is cross-referenced by the API, UI, and test contracts. It contains 19 functional requirements, 26 business rules, 24 acceptance criteria, and a Definition of Done. The contract-first document was delivered in PR #58 before the implementation PRs were completed.

| Contract area     | Selected decision                                                                               | Traceability source                   |
|-------------------|-------------------------------------------------------------------------------------------------|---------------------------------------|
| Requester context | Development Requester selection is a test context, not authentication.                          | FR-01–FR-04, BR-03–BR-07, AC-01–AC-03 |
| Ticket creation   | The server owns Ticket Number, Ticket Date, requester ownership, and initial <b>NEW</b> status. | FR-06–FR-08, BR-01–BR-13, AC-05–AC-10 |
| Ownership         | List, detail, and attachment operations are scoped to the selected Requester.                   | FR-09–FR-16, BR-23, AC-11–AC-21       |
| Scope boundary    | Authentication, IT Staff workflow, collaboration, and later status transitions remain excluded. | Scope and DoD sections                |

### Rendered contract evidence

The source contract is complete and implemented on **lab2-staging**. The final report still needs a readable rendered copy and one screenshot proving that the specification existed before the main implementation PRs were completed.

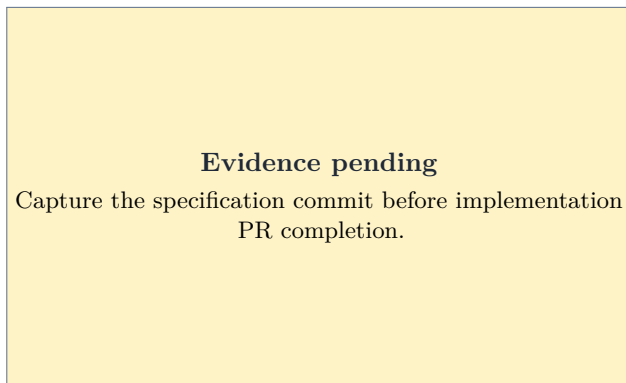


Figure 5: Specification-before-implementation proof, associated with the contract PR.

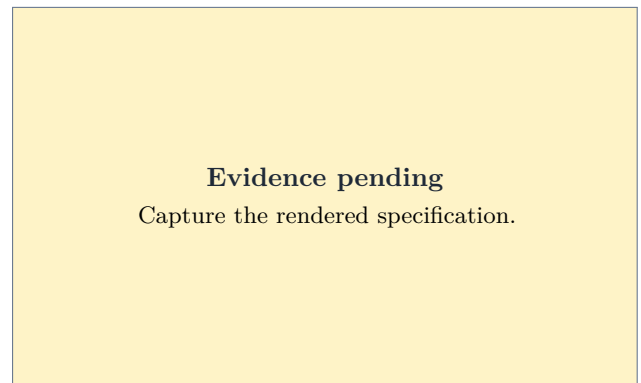


Figure 6: Rendered **specification.md** with numbered FR, BR, AC, and DoD sections.

## 3 Answer Part 3: Test DD and Traceability

The planned matrix and acceptance-criteria mapping are maintained in [docs/lab-02/tests.md](#). The current integrated staging validation record contains the following results:

| Boundary              | Result                                     | Main evidence path                                                      |
|-----------------------|--------------------------------------------|-------------------------------------------------------------------------|
| Client unit/UI suite  | 8 files, 32 tests passed                   | <code>client/tests/lab-01/</code> and <code>client/tests/lab-02/</code> |
| Server unit/API suite | 13 files, 70 tests passed                  | <code>server/tests/lab-01/</code> and <code>server/tests/lab-02/</code> |
| Combined test command | 21 files, 102 tests passed                 | <code>bun run test</code>                                               |
| Browser E2E           | 1 non-skipped test and 36 screenshots      | <code>e2e/lab-02/requester-ticket-flow.spec.ts</code>                   |
| Full repository gate  | Passed on the integrated staging base-line | <code>bun run verify</code> ; final-main rerun pending                  |

The principal Lab 2 test paths are:

- `server/tests/lab-02/create-ticket.api.test.ts`, `server/tests/lab-02/my-tickets.api.test.ts`, `server/tests/lab-02/ticket-detail.api.test.ts`, and `server/tests/lab-02/attachments.api.test.ts`;
- `client/tests/lab-02/CreateTicket.test.tsx`, `client/tests/lab-02/MyTickets.test.tsx`, `client/tests/lab-02/RequesterTicketDetail.test.tsx`, and `client/tests/lab-02/AttachmentSection.test.tsx`.

Every AC is mapped to at least one test ID in the Markdown matrix.

The 2026-09-03 `bun run verify` run passed on the report branch above the `lab2-staging` product baseline. It checked Biome, Lefthook, Prisma, type checking, 21 test files with 102 tests, and both workspace builds. The labsheet requires complete unit, API, and UI output from final `main`; that evidence cannot be claimed until the release is merged and rerun.

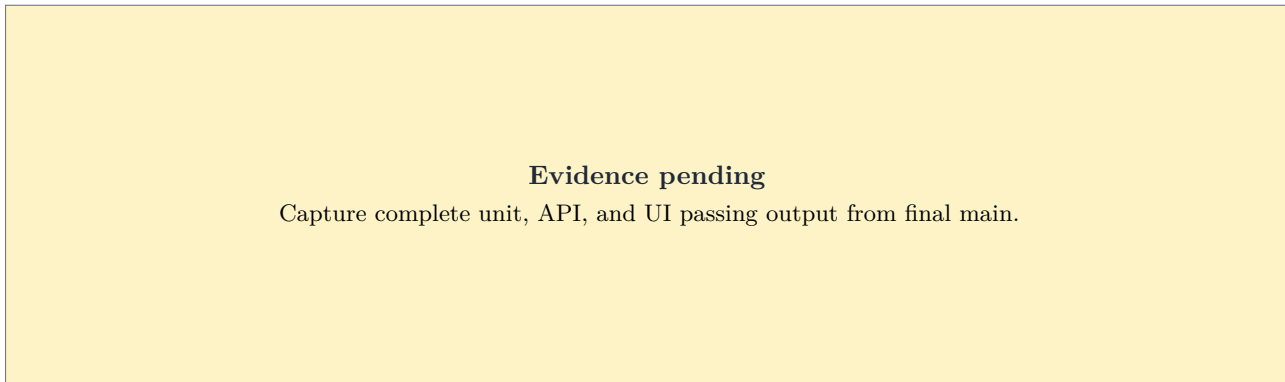


Figure 7: Final-main test output required by the labsheet.

## 4 Answer Part 4: AI Use with Reflection

The AI-use record is linked at [docs/lab-02/ai-use.md](#). The implementation assistant was OpenAI Codex using GPT-5. The student reviewed the PDF-derived contracts, generated changes, tests, screenshots, and review responses.

| ID | Selected prompt focus                                                          | Effect                                                                                 |
|----|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| P1 | Read the Lab 2 PDF and convert workflows and submission parts into a contract. | Established numbered FR, BR, AC, and DoD traceability.                                 |
| P2 | Create the Issue and branch plan from the contract.                            | Kept one feature branch per Lab 2 Issue and preserved the staging flow.                |
| P3 | Implement data, seed, requester context, API, and ownership boundaries.        | Added the requester-facing backend contract without authentication or later-lab scope. |
| P4 | Implement Create Ticket, My Tickets, Detail, and attachments.                  | Connected the four product areas to the approved API and UI decisions.                 |
| P5 | Add tests with explicit FR/BR/AC traceability.                                 | Produced unit, API, client UI, responsive, and E2E coverage.                           |
| P6 | Inspect the requester flow at desktop, tablet, and mobile sizes.               | Generated the responsive screenshot inventory and overflow assertions.                 |
| P7 | Review the complete Lab 2 PR sequence and address comments.                    | Updated implementation, tests, evidence, and reviewer records after human feedback.    |
| P8 | Check report readiness against Answer Parts 1-9.                               | Separated staging evidence from final-main and missing-capture work.                   |

### My Reflection

AI assistance was most useful for translating the labsheet into small, traceable contracts and checking the relationships between implementation, tests, evidence, and review records. Human responsibility remained essential: the student confirmed scope, reviewed generated changes, inspected screenshots, validated the environment, and did not treat automated review output as peer approval.

## 5 Answer Part 5: Development Requester Select Screen

Lab 2 uses a simulated login screen to select an active Development Requester for the session. It is explicitly not authentication. Inactive seeded Requesters are excluded, Continue is disabled until a valid active selection is made, and Change Requester clears the previous requester-specific view.

**Evidence pending**

Capture the Development Requester selection screen, active dropdown, selected-user display, loading/failure state, and Change Requester action.

Figure 8: Development Requester selection screen required for the final report.

The current client tests and E2E flow exercise these states, but the current 36-file screenshot inventory does not contain a dedicated requester-selection capture. This evidence must be added before the final PDF.

## 6 Answer Part 6: Working Ticket Screen - Create Mode

Create Ticket accepts Category, Related System, Requested Priority, Summary, Description, and permitted attachments. The selected requester is displayed as read-only context. The server generates the official Ticket Number, Ticket Date, requester ownership, and initial NEW status. Attachment upload is a separate operation so a saved Ticket remains visible if an upload fails.

**Evidence pending**

Capture requester context before entering Create Ticket.

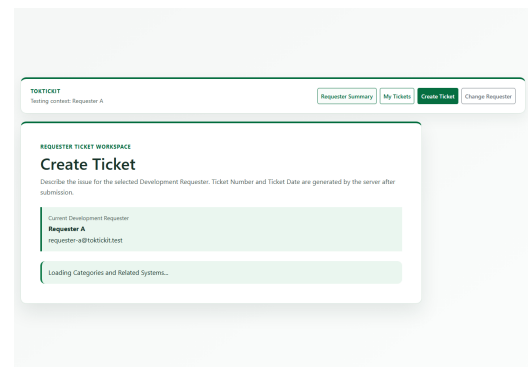


Figure 10: Reference data loading from the API.

Figure 9: Requester selection and matching context.

Capture the initial empty Create Ticket form.

Figure 11: Initial editable form with required indicators.

Capture the disabled busy Submit state.

Figure 13: Submitting state and duplicate-submit protection.

[illegible]

Figure 12: Field-level messages for an invalid submission.

[illegible]

Figure 14: Backend-generated Ticket Number and saved values.

[illegible]

Figure 15: Safe API failure with entered values preserved.

Capture Create Ticket with one valid and one invalid attachment.

Figure 16: Attachment policy feedback in Create mode.

The current UI/API tests and E2E evidence support the following contract claims:

- API-02 and E2E-01 verify a successful create response, saved Ticket, backend Ticket Number, and requester context.
- UI-04 and API-03 verify field-level validation and no invalid Ticket persistence.
- UI-05 and API-05 verify busy-submit protection, preserved values, and separate attachment failure behavior.
- The final report still needs readable evidence of the saved `requesterId` and database values, not only the UI success message.

## 7 Answer Part 7: Working My Tickets Screen

My Tickets is always scoped by the selected Development Requester. It supports search over Ticket Number and Summary, Category/Related System/Priority/Status filters, deterministic sorting, one-based pagination, empty and no-results states, and recoverable API failure.

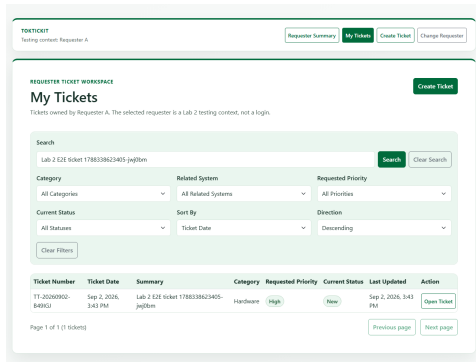


Figure 17: Requester-owned results with list controls.

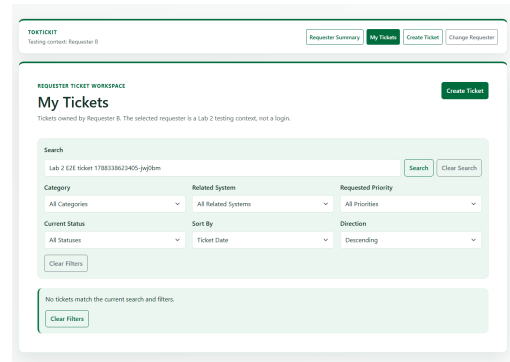


Figure 18: A different Requester cannot see the original Requester's Tickets.

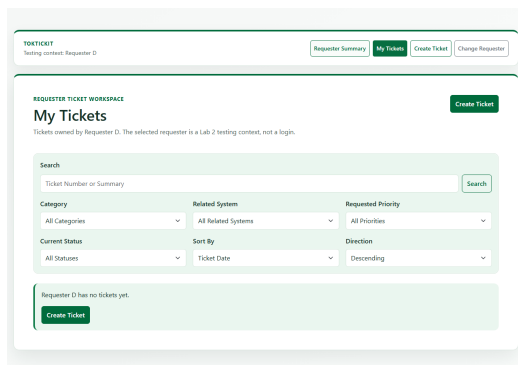


Figure 19: No Tickets for the current Requester.

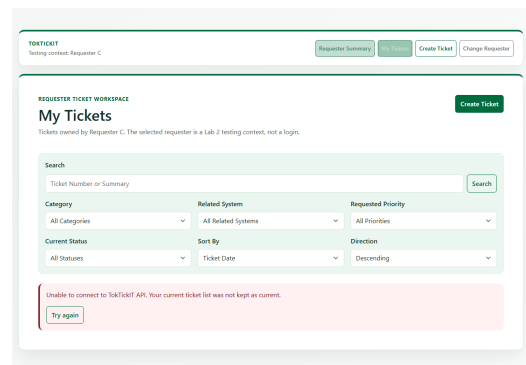


Figure 20: Recoverable list failure.

The client and server tests verify search, filters, sorting, pagination, ownership isolation, empty/no-results distinction, stale-response protection, and API failure. The final report should add a clearly identifiable capture showing a non-default filter, sort direction, and page transition together; the current filtered screenshot is useful evidence but does not prove every query control separately.

## 8 Answer Part 8: Working Ticket Screen - View Mode and Attachments

Owned Ticket Detail renders Ticket fields read-only and separates the Attachment section. Active attachments can be uploaded and downloaded. Removal requires confirmation and a 3-200 character reason, retains metadata, and returns 410 ATTACHMENT\_REMOVED when a removed file is requested. Cross-Requester Ticket and Attachment access is rejected safely.

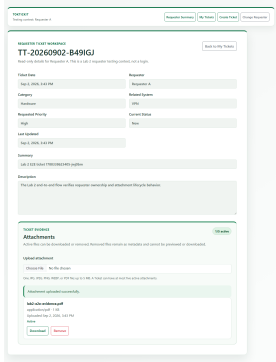


Figure 21: Read-only Ticket Detail with active attachment.

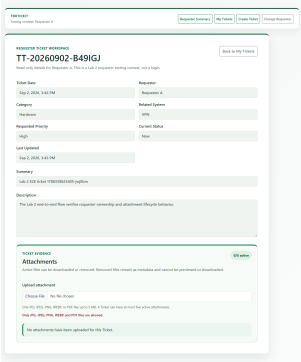


Figure 22: Unsupported attachment feedback.

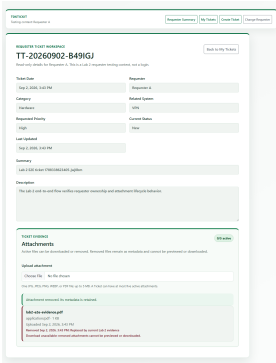


Figure 23: Retained removed metadata and reason.

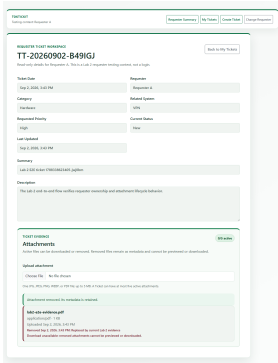


Figure 24: Removed content is not downloadable or pre-viewable.

API-09 through API-16 and E2E-02 cover owned detail, attachment metadata, upload, active download, soft removal, repeat removal, invalid files, storage errors, removed-download rejection, and cross-owner access. The final report should include a readable test-output or API excerpt for the unauthorized Ticket/Attachment requests in addition to the UI screenshots.

Evidence pending

Capture the API test output showing a different Requester receives the safe ownership rejection.

Figure 25: Unauthorized Ticket and Attachment access evidence required for Part 8.

## 9 Answer Part 9: Zen Green UI and Responsive Evidence

The complete visual contract is linked at [docs/lab-02/ui-spec.md](#). The integrated E2E flow checks 1280x900, 820x900, and 390x844 viewports, and the current inventory contains 36 captures across 12 state prefixes.

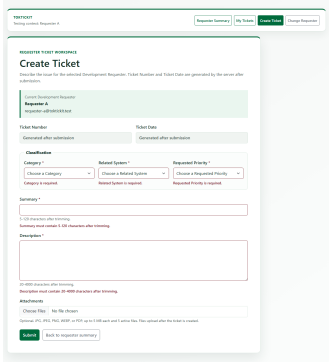


Figure 26: Desktop validation.

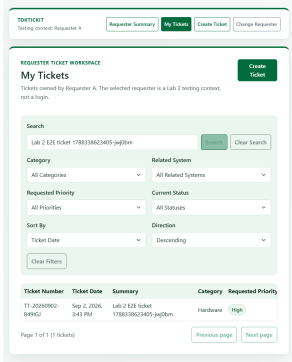


Figure 27: Tablet list controls.



Figure 28: Mobile blocked-download state.

| Visual checklist item                      | Integrated evidence                   | Final report status                                         |
|--------------------------------------------|---------------------------------------|-------------------------------------------------------------|
| Zen Green colors and shared components     | Client contract tests and screenshots | Rendered <code>ui-spec.md</code> and human sign-off pending |
| Editable/read-only fields                  | Create and Detail UI tests            | Representative screenshots available                        |
| Validation placement and state text        | Client tests and validation captures  | Representative evidence available                           |
| Button hierarchy and disabled busy state   | Client tests and UI contract          | Dedicated submitting capture pending                        |
| Clipping, overlap, and horizontal overflow | E2E viewport assertions               | Final-main rerun and visual sign-off pending                |

The final report must include the completed visual checklist and all readable desktop, tablet, and mobile evidence. The current screenshot set is a strong integrated staging baseline, but it is not yet the complete final-main submission package.